

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
3	(a)	(i)	Use of the equation $\frac{M_2 d}{(M_1 + M_2)} = x$ even if wrong M_2 (1) $\frac{9.2 \times 10^{24} \times 5.3 \times 10^{10}}{(9.5 \times 10^{24} + 9.2 \times 10^{24})}$ seen (1) ({ $+9.2 \times 10^{24}$ } can be omitted from the above equation) or 513 263 [m] seen etc.	1 1			2	2	
		(ii)	$v = \frac{2\pi r}{T}$ applied or $v = \omega r$ and $\omega = \frac{2\pi}{T}$ combined (1) Conversion of 130 days ($130 \times 24 \times 3600$) (1) Answer = $0.287 \text{ [m s}^{-1}\text{]}$ (ecf on day conversion) (1)	1	1 1		3	3	
		(iii)	Redshift is small or the redshift is proportional to velocity or small $\Delta\lambda$ or shown using the Doppler equation (1) Small wavelength change is difficult to measure or shifted wavelength is too close to the original (1)		2		2		
	(b)	(i)	$F = \frac{GMm}{r^2}$ used (1) Answer = $2.08 \times 10^{23} \text{ [N]}$ (1)	1	1		2	2	
		(ii)	Method for obtaining stress i.e. $\frac{F}{A}$ (1) $1.35 \times 10^9 \text{ [Pa]}$ (1) Correct conclusion e.g. the steel bar would break ecf (1) Theory doesn't work / Newton's grav law used successfully for centuries / won't work for elliptical orbits / would melt (1) Alternative: Use of $F = \sigma A$ (1) $F = 6.9 \times 10^{22} \text{ [N]}$ (1)			4	4	2	

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	(c)	(i)	$PE = -\frac{GMm}{r}$ used N.B. ecf if $PE = Fr$ used (1) Answer = $[-]1.10 \times 10^{34}$ [J] (1)	1	1		2	2	
		(ii)	Realising KE is involved (1) Correct variation with distance for KE or velocity or as PE increases KE decreases or converse (1)			2	2		
			Question 3 total	5	6	6	17	11	0